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Metal-organic framework MIL-100(Fe) for dye removal in aqueous solutions: Prediction by artificial neural network and response surface methodology modeling☆

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ABSTRACT

In this study, a metal organic framework MIL-100(Fe) was synthesized for rhodamine B (RB) removal from aqueous solutions. An experimental design was conducted using a central composite design (CCD) method to obtain the RB adsorption data (n = 30) from batch experiments. In the CCD approach, solution pH, adsorbent dose, and initial RB concentration were included as input variables, whereas RB removal rate was employed as an output variable. Response surface methodology (RSM) and artificial neural network (ANN) modeling were performed using the adsorption data. In RSM modeling, the cubic regression model was developed, which was adequate to describe the RB adsorption according to analysis of variance. Meanwhile, the ANN model with the topology of 3:8:1 (three input variables, eight neurons in one hidden layer, and one output variable) was developed. In order to further compare the performance between the RSM and ANN models, additional adsorption data (n = 8) were produced under experimental conditions, which were randomly selected in the range of the input variables employed in the CCD matrix. The analysis showed that the ANN model (R^2 = 0.821) had better predictability than the RSM model (R^2 = 0.733) for the RB removal rate. Based on the ANN model, the optimum RB removal rate (>99.9%) was predicted at pH 5.3, adsorbent dose 2.0 g L^{-1}, and initial RB concentration 73 mg L^{-1}. In addition, pH was determined to be the most important input variable affecting the RB removal rate. This study demonstrated that the ANN model could be successfully employed to model and optimize RB adsorption to the MIL-100(Fe).

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1. Introduction

Metal-organic frameworks (MOFs) are a new class of inorganic-organic hybrid materials with open crystalline networks and porosity. In the synthesis of MOFs, strong coordination bonds are formed between organic linkers and metal-containing nodes to produce crystalline materials (Feng et al., 2018; Kökçam-Demir et al., 2020). MOFs have attracted great attention from many researchers because of the easy tunability in their shape and pore size by employing various organic linkers and metal ions (Zhou and Kitagawa, 2014). MOFs have been widely used in industrial fields including chemical catalysis, fuel cells, hydrogen storage, sensing, and supercapacitor (Furukawa et al., 2013; Tehrani et al., 2017). Recently, water-stable MOFs have been applied for contaminant removal from aqueous solutions (Cai et al., 2016; Seo et al., 2016; Wang et al., 2017; Zhan et al., 2018; Zhang et al., 2019; Esrafili et al., 2019a, 2019b). Furthermore, some researchers have synthesized water stable MOF-based composites with carbon nanotubes, graphene, and metal nanoparticles for application to water and wastewater treatment (Torad et al., 2014; Aslam et al., 2017; Liu et al., 2018; Mahmoodi et al., 2019).

Dye wastewater pollution is a serious environmental problem worldwide. Dyes are extensively used in various industries such as cosmetics, textiles, paints, plastics, leather, and tanneries. A large

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amount of wastewater is generated from these industries and discharged into water bodies, posing a great threat to the aqueous environment and human health (Zhu et al., 2016; Fan et al., 2018). Various physical, chemical, and biological technologies have been applied to remove dyes from wastewater, including adsorption, anaerobic/aerobic biological processes, chemical oxidation, coagulation, membrane filtration, and photocatalysis (Adeyemo et al., 2012). Among the treatment technologies, adsorption is widely applied for the removal of dyes in wastewater due to its simplistic operation and cost-effectiveness (Haque et al., 2011; Zhu et al., 2016; Fan et al., 2018). Recently, MOF-based adsorbents have been tested for the removal of various dyes, including methyl orange, methylene blue, methyl red, rhodamine 6G, rhodamine B (RB), crystal violet, and acid orange 10 in aqueous solutions (Haque et al., 2011; Huo and Yan, 2012; Tan et al., 2015; Zhao et al., 2015; Yilmaz et al., 2016; Duan et al., 2016; Liu et al., 2016; Ayati et al., 2016; Zhu et al., 2016; Aslam et al., 2017; Abbasi et al., 2017; Wei et al., 2018; Fan et al., 2018). These studies showed that MOFs are promising porous nanostructured adsorbents for dye removal in wastewater.

In the application of the MOF-based adsorbents for dye removal, multi-parameter modeling approaches such as response surface methodology (RSM) and artificial neural network (ANN) can be used to predict and optimize the adsorption process. RSM is a mathematical and statistical technique that is employed to model the effects of various input parameters and their interactions on the output variable (response of interest) and to obtain optimal conditions for the response variable (Jahed Armaghani et al., 2014; Apostolopoulou et al., 2020). In the RSM approach, a parametric model is developed based on the polynomial equation to optimize the adsorption process variables and to predict the target response (Sharma et al., 2015; Sharafi et al., 2019; Samadi-Maybodi and Nikou, 2020). ANN is a modeling technique that imitates the human neural system to process self-learning. The ANN architecture is composed of an input layer, hidden layer, and output layer along with neurons to model the nonlinear complex systems. In the ANN approach, a non-parametric model, which is expressed with transfer functions, weights, and biases, is developed for prediction of the adsorption process (Esfandiari et al., 2017; Gadekar and Ahammed, 2019). Thus far, very few studies have been reported for the application of the RSM and/or ANN modeling approaches to optimize the dye removal by the MOF-based adsorbents, including the ANN modeling for dye (fast green, eosin yellow (EY), and quinine yellow) removal by the MOF-based hybrid adsorbent (AC-MOF-5) (Askari et al., 2017), the ANN modeling for adsorption of malachite green (MG) and EY to the monolithic HKUST-1 MOF (Parsazadeh et al., 2018), and the RSM and ANN modeling for removal of MG and RB by the MOF-based nanocomposites (NH_2-CNT/Fe_2O_3/ZIF-8) (Mahmoodi et al., 2019). Further studies are needed to improve knowledge of optimizing the dye adsorption to the MOF-based adsorbents with the RSM and ANN modeling.

In this study, MIL-100(Fe) was synthesized for dye removal from aqueous solutions. MIL-100(Fe) was selected as a MOF-based adsorbent because it had environmental-friendly nature and higher water stability than other MOF-based adsorbents (Fang et al., 2018; Nehra et al., 2019; Zhang et al., 2019). RB was used as a target dye pollutant because it was widely used in the textile industry. RB has a xanthene core (Fig. S1, supplementary information, SI), with a chemical formula of $CH_2[C_6H_4]_2O$ and tends to be fluorescent (Baldev et al., 2013). Based on the literature review, this study focused on four parts: (i) experiments for synthesized MIL-100(Fe) toward RB adsorption with a central composite design (CCD), (ii) model development for RB adsorption to the MIL-100(Fe) using RSM and ANN approaches, (iii) comparison of the performance between the RSM and ANN models with additional adsorption data, and (iv) optimization of RB adsorption to the MIL-100(Fe) by developed RSM or ANN model.

2. Materials and methods

2.1. Synthesis of MIL-100(Fe)

Iron(II) chloride tetrahydrate ($FeCl_2 \cdot 4H_2O$, 99%) and trimesic acid (H_3BTC, 95%) were purchased from Sigma-Aldrich (Saint Louis, MO, USA). Sodium hydroxide (NaOH) and ethyl alcohol (C_2H_5OH, 94.5%) were obtained from Duksan Pure Chemicals (Ansan, Republic of Korea) and Daejung Chemicals & Metals (Siheung, Republic of Korea), respectively. The MIL-100(Fe) was synthesized following the procedures described in Guesh et al. (2017), who manufactured the MIL-100(Fe) at room temperature in the absence of corrosive acid. Briefly, NaOH (24.0 mM) and H_3BTC (8.0 mM) were dissolved in 30 mL of deionized water (DW) (solution 1). In addition, $FeCl_2 \cdot 4H_2O$ (12.07 mM) was dissolved in 96 mL of DW (solution 2). Solution (1) was added dropwise to solution (2), stirred for 24 h at room temperature, and centrifuged at 4000 rpm to obtain the solid product. Then, the solid was washed three times with DW and once more with ethyl alcohol, and dried overnight at 90 °C in a vacuum drying oven (SH-VDO-08NG, Samheung, Seoul, Republic of Korea) to obtain the final MIL-100(Fe).

2.2. Characterization of MIL-100(Fe)

The physical and chemical characteristics of the MIL-100(Fe) were analyzed using various instruments. Nitrogen gas (N_2) adsorption-desorption analysis was conducted using a surface area analyzer (BELSORP-MAX, MicrotracBEL, Osaka, Japan) to determine the Brunauer-Emmett-Teller (BET) surface area and pore diameter. An electrophoretic light scattering (ELS) spectrophotometer (ELSZ-1000, Otsuka Electronics, Tokyo, Japan) was employed to measure the zeta potential and particle diameter. A thermogravimetric analyzer (Discovery TGA, TA Instruments, New Castle, DE, USA) was used for thermogravimetric analysis (TGA) by heating the MIL-100(Fe) to 800 °C with a heating rate of 10 °C min^{-1}.

Furthermore, a field emission scanning electron microscope (FESEM, Supra 55VP, Carl Zeiss, Oberkochen, Germany) was employed to examine the surface morphology. An energy dispersive X-ray spectroscopy (EDS) pattern was determined using FESEM. The X-ray diffraction (XRD) pattern was obtained using powder X-ray diffractometry (D8 Advance, Bruker, Billerica, MA, USA) with the 2θ range from 2° to 20°. Fourier transform infrared (FTIR) analysis was conducted using a spectrophotometer (Nicolet 6700, Thermo Scientific, Waltham, MA, USA) to determine the infrared spectra.

Water stability of the MIL-100(Fe) was analyzed in deionized water (DW) and RB solution. The stability tests were performed in duplicate at the solution pHs of 2, 7, and 12.80 mg of the MIL-100(Fe) was added into 40 mL of DW and RB solution (100 mg L^{-1}) in polypropylene conical tubes and stirred in a shaking incubator (Daihan Science, Seoul, Republic of Korea) at 150 rpm and 30 °C for 24 h. After reaction, the amount of Fe leaching from the MIL-100(Fe) was quantified using Inductively Coupled Plasma-Optical Emission Spectrometry (ICP-OES) (iCAP 7000, Thermo Fisher Scientific, Waltham, MA, USA).

2.3. Adsorption experiments

Batch experiments were performed for RB adsorption to the MIL-100(Fe). A dye stock solution (500 mg L^{-1}) was prepared from RB (Sigma-Aldrich, Saint Louis, MO, USA). Adsorption experiments were conducted with polypropylene conical tubes containing

50 mL of the RB solution (initial concentration = 50 mg L^{-1}, pH = 4) and MIL-100(Fe) (adsorbent dose = 1 g L^{-1}), unless stated otherwise. The duplicate samples were stirred in a shaking incubator at 150 rpm and 30 °C for 24 h. After reaction, the samples were collected and filtered through a 0.45-μm membrane filter. The concentrations of RB in the filtrate were determined using a UV–Vis spectrophotometer (Genesys 10S, Thermo Scientific, Waltham, MA, USA) at a wavelength of 470 nm based on the calibration curve obtained from the spectra of the RB standard solutions.

2.4. Experimental design and modeling analysis

2.4.1. Central composite design

A schematic diagram of the experimental design and modeling analysis for RB adsorption to the MIL-100(Fe) is illustrated in Fig. 1. First, an experimental design was performed using the CCD method. The number of experimental points ($N = 15$) were calculated using the following formula (Nair et al., 2014):

$$N = 2^k + 2k + cp \tag{1}$$

where k is the number of input variables (= 3), $2^k$is the number of factorial points (= 8), $2k$ is the number of axial points (= 6), and cp is the number of center points (= 1). Experimental range and level of input variable in the CCD approach are presented in Table 1. In this study, input variables including pH (A), adsorbent dose (B), and initial RB concentration (C) were chosen as five levels and labeled as −2 to +2. The response parameter was the removal rate of RB. The range of input variables was determined based on the preliminary adsorption experiments described in Section 2.3. The CCD matrix in Table 2 was obtained using the Design Expert software (version 12, Stat-Ease, Minneapolis, MN, USA). Based on the CCD matrix, adsorption experiments (n = 30) were conducted in duplicate for each experimental point (16 factorial points, 12 axial points, and 2 center points).

Table 1
Experimental range and level of input variables in CCD ($\alpha = 2$).

Input variable	Symbol	Level				
		-α	−1	0	+1	+α
pH	A	2	3	4	5	6
Adsorbent dose (g L^{-1})	B	1.0	1.5	2.0	2.5	3.0
Initial RB conc. (mg L^{-1})	C	50	75	100	125	150

2.4.2. Response surface methodology

RSM was applied to analyze the RB adsorption data obtained from the designed experiments. The influence of the input variables on the RB removal rate was modeled by the following cubic polynomial equation (Saldana-Robles et al., 2014; Boudechiche et al., 2017):

$$y = \beta_0 + \sum_{i=1}^{k} \beta_i x_i + \sum_{i=1}^{k} \beta_{ii} x_{ii}^2 + \sum_{i=1}^{k} \beta_{iii} x_{iii}^3 + \sum_{i=1}^{k-1} \sum_{j=i+1}^{k} \beta_{ij} x_i x_j + \sum_{i=1}^{k-2} \times \sum_{i=j+1}^{k-1} \sum_{k=j+1}^{k} \beta_{ijk} x_i x_j x_k + \varepsilon \tag{2}$$

where y is the output value (RB removal rate), x_i, x_j, and $x_k$are the input variables, β_0, β_i, β_{ii}, and β_{iii} are the regression coefficients for intercept, linear, quadratic, and cubic term, respectively, β_{ij} and β_{ijk} are the cross-product term coefficients, ε is the error associated

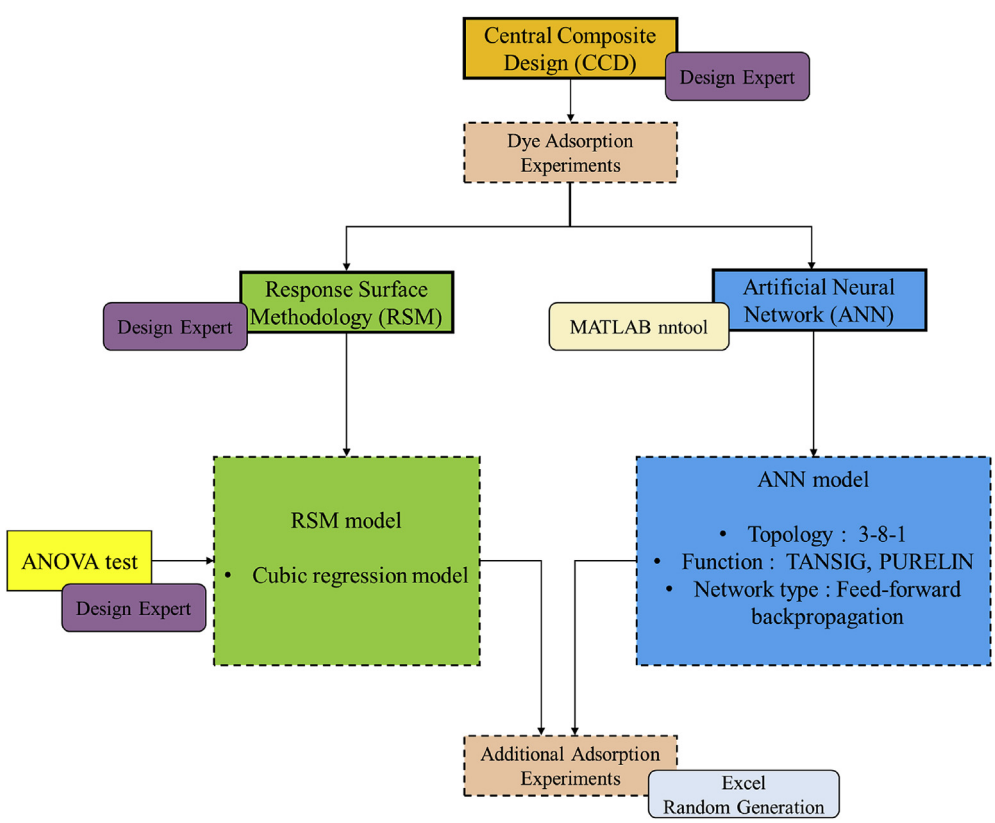

Fig. 1. Schematic diagram of experimental design and modeling analysis.

Table 2
CCD matrix, observed removal rates, and predicted values from RSM and ANN modeling.

Exp.	Input variable			Removal rate (%)		
	A (pH)	B (adsorbent dose)	C (initial RB conc.)	Observed	Predicted-RSM	Predicted-ANN
1	2	2.0	100	48.9	47.7	48.4
2	3	1.5	75	86.6	86.4	86.6
3	3	1.5	125	49.6	52.8	52.1
4	3	2.5	75	90.7	91.9	90.7
5	3	2.5	125	75.2	74.4	73.8
6	4	1.0	100	52.0	54.1	57.5
7	4	2.0	50	94.7	94.6	95.3
8	4	2.0	150	82.3	79.7	80.3
9	4	3.0	100	92.9	93.3	86.6
10	5	1.5	75	86.9	85.2	91.0
11	5	1.5	125	60.4	60.0	60.4
12	5	2.5	75	95.6	96.4	95.9
13	5	2.5	125	75.2	79.1	75.2
14	6	2.0	100	90.0	88.4	89.0
15	4	2.0	100	91.3	84.3	85.6
16	2	2.0	100	47.9	47.7	48.4
17	3	1.5	75	85.0	86.4	86.6
18	3	1.5	125	54.6	52.8	52.1
19	3	2.5	75	91.8	91.9	90.7
20	3	2.5	125	72.3	74.4	73.8
21	4	1.0	100	57.5	54.1	57.5
22	4	2.0	50	95.9	94.6	95.3
23	4	2.0	150	78.4	79.7	80.3
24	4	3.0	100	95.0	93.3	86.6
25	5	1.5	75	82.2	85.2	91.0
26	5	1.5	125	58.4	60.0	60.4
27	5	2.5	75	95.9	96.4	95.9
28	5	2.5	125	81.7	79.1	75.2
29	6	2.0	100	88.1	88.4	89.0
30	4	2.0	100	79.9	84.3	85.6

with the adsorption experiments, and k is the number of input variables (= 3). The input variables were normalized (x_n) in the range of −1 to +1 to avoid the scaling effect using the following normalization formula:

$$x_n = 2 \times \frac{(x - x_{min})}{(x_{max} - x_{min})} - 1 \tag{3}$$

where x is the input value, x_{max} and x_{min} are the maximum and minimum value of x, respectively. The RSM model was developed based on equation (2) using the Design Expert software. The analysis of variance (ANOVA) test was performed to evaluate the fitness of the developed RSM model to the RB adsorption data based on the Fisher value (F-value), probability value (P-value), and coefficient of determination value (R^2):

$$R^2 = 1 - \frac{\sum_{i=1}^{N}\left(y_{i,\,pre} - y_{i,\,obs}\right)^2}{\sum_{i=1}^{N}\left(y_{i,\,obs} - y_{ave}\right)^2} \tag{4}$$

where $y_{i,pre}$, $y_{i,obs}$, and y_{ave} are the predicted value, observed value, and average value of RB removal rate, respectively.

2.4.3. Artificial neural network

ANN was also used to model the RB adsorption data from the designed experiments. The ANN architecture for the RB adsorption is illustrated in Fig. S2 (SI). The ANN model was composed of three layers (input, hidden, and output layers). The input layer could include various input variables, whereas the hidden and output layers could contain a number of neurons for each layer (Jahed Armaghani et al., 2017; Mahmoodi et al., 2019). In the ANN modeling, the input variables were also normalized in the range of −1 to 1. The effect of the input variables on the RB removal rate was modeled by the following ANN model (Ghaedi et al., 2015; Baziar et al., 2017):

$$y = \text{Purelin}\left[\boldsymbol{w}_O \times \text{Tansig}\left(\boldsymbol{w}_H \times \begin{bmatrix} A \\ B \\ C \end{bmatrix} + \boldsymbol{b}_H\right) + \boldsymbol{b}_O\right] \tag{5}$$

where $\boldsymbol{w}_H$ and $\boldsymbol{w}_O$ are the weight values of hidden layer and output layer, respectively, and $\boldsymbol{b}_H$ and $\boldsymbol{b}_O$ are the bias values of hidden layer and output layer, respectively. The hyperbolic tangent sigmoid transfer function (Tansig) was used as a transfer function for the hidden layer, whereas the linear transfer function (Purelin) was used for the output layer. The ANN model was developed based on equation (5) using the MATLAB nntool (version R2019b, MathWorks, Natick, MA, USA). The mean squared error (MSE) was used as an error function:

$$\text{MSE} = \frac{1}{N}\sum_{i=1}^{N}\left(\left|y_{i,\,pre} - y_{i,\,obs}\right|\right)^2 \tag{6}$$

The Levenberg-Marquardt algorithm was selected for the modeling process because it could result in the smaller MSE value (Baziar et al., 2017). In the ANN model development, a total of 30 data points (Table 2) were randomly divided into three subsets including training set (60%), validating set (20%), and testing set (20%). Eighteen data points were used in the training phase where weight and bias values were modified and adjusted so that the error between the observed and predicted outputs could be reduced (Ghaedi et al., 2017; Gedekar et al., 2019). In the validating phase, six data points were used to prevent overfitting (Ma et al., 2019). In the testing phase, six data points were used to evaluate the fitness of the developed ANN model to the experimental data.

3. Results and discussion

3.1. Characteristics of MIL-100(Fe)

The FESEM and digital images of the MIL-100(Fe) are presented in Fig. 2. The synthesized MIL-100(Fe) was a brownish particle with an octahedral shape (Fig. 2a). The physical characteristics of the MIL-100(Fe) are summarized in Table S1 (SI). According to ELS spectrophotometer analysis, the MIL-100(Fe) had an average hydro-dynamic diameter of 4.3 ± 0.9 μm with a point of zero charge (pH_{pzc}) of 4.2 (see Fig. 2b). The N_2 adsorption-desorption analysis is illustrated in Fig. S3 (SI). The MIL-100(Fe) had BET surface area of 1752.2 $m^2\ g^{-1}$, total pore volume of 0.730 $cm^3\ g^{-1}$, and average pore diameter of 1.67 nm (Table S1, SI).

The XRD pattern and FTIR spectrum of the MIL-100(Fe) are presented in Fig. S4 (SI). In the XRD pattern (Fig. S4a), the characteristic peaks at 3.4, 4.0, 4.8, 6.3, and 11.0° corresponded to the (022), (113), (004), (333), and (428) planes of the MIL-100(Fe). Our XRD pattern was in accordance with the previously reported XRD pattern of MIL-100(Fe)-RT 24 h, which was synthesized and washed at room temperature by Guesh et al. (2017). In the FTIR spectrum of the MIL-100(Fe) (Fig. S4b), O—H stretching was detected at 2972 cm^{-1} (Mohammadifard et al., 2019), whereas C=O stretching was detected at 1700 and 1620 cm^{-1} (Duan et al., 2016; Lee et al., 2017; Mohammadifard et al., 2019). The peak at 1365 cm^{-1} corresponded to the C=C bond, while the peaks at 759 and 708 cm^{-1} were assigned to Fe—OH vibration (Lee et al., 2017; Mohammadifard et al., 2019).

The EDS pattern of the MIL-100(Fe) in Fig. S5 (SI) indicated that the weight percentages of carbon (C), oxygen (O), and iron (Fe)

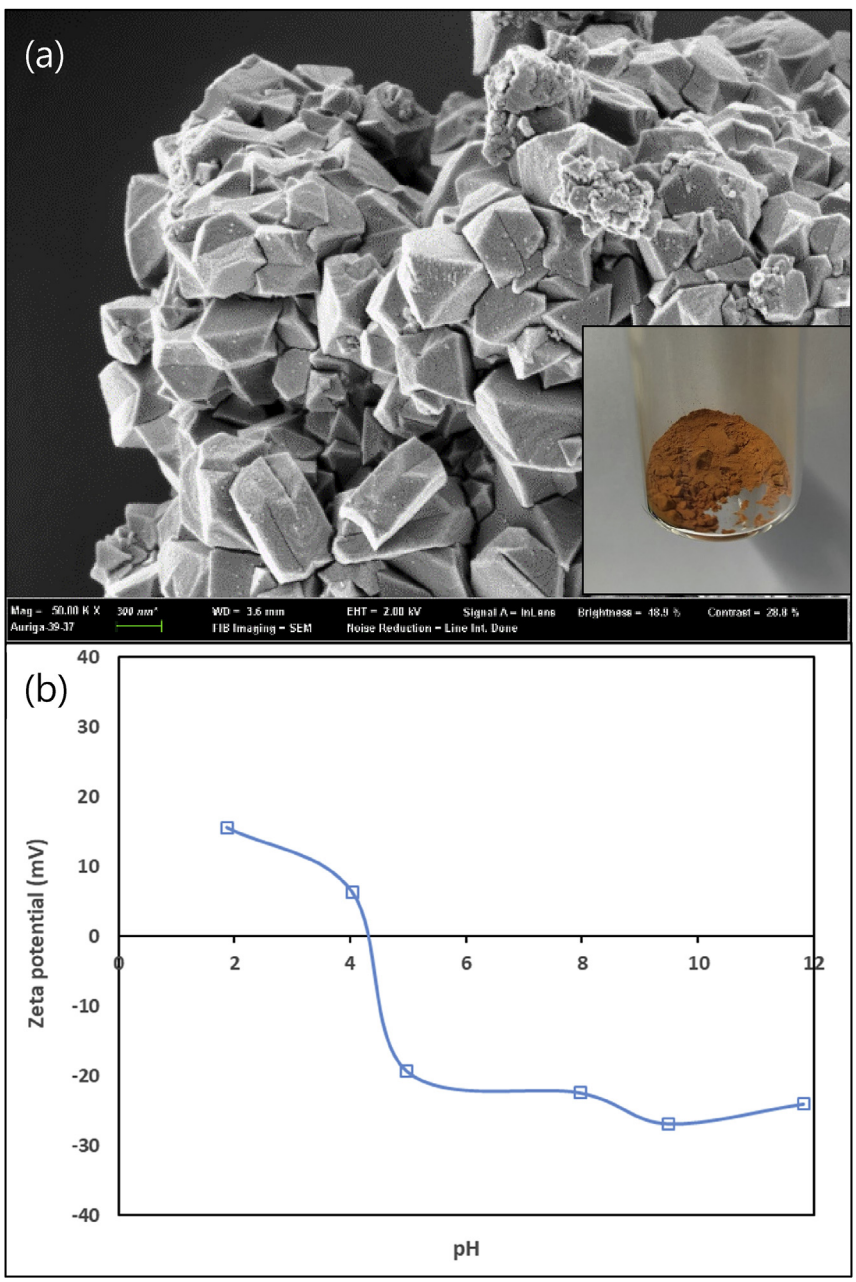

Fig. 2. FESEM and digital images of the MIL-100(Fe) synthesized in this study.

were 47.3, 34.6, and 18.1%, respectively. The presence of iron (Fe) was confirmed with L_α, K_α, and K_β peaks at 0.705, 6.403, and 7.057 keV, respectively. The TGA in Fig. S6 (SI) displayed a three-step mass loss of the MIL-100(Fe) in the temperature range of 20–500 °C. In the first range (20–85 °C), the mass loss (15.1%) occurred due to desorption of water molecules inside the pore of the MIL-100(Fe). The mass loss (15.4%) in the second range of 85–360 °C was related to the evaporation of water molecules interacted with the Fe trimers. In the range of 360–500 °C, the mass loss of 44.7% was ascribed to the combustion of trimesic acid and collapse of the framework (Duan et al., 2016; Mohammadifard et al., 2019). The TGA showed that the synthesized MIL-100(Fe) had an acceptable thermal stability up to 360 °C. The water stability of the MIL-100(Fe) in DW and RB solutions are presented in Table S2 (SI). A negligible amount of Fe was leached from the MIL-100(Fe) in the different pH solutions, indicating that the MIL-100(Fe) had great water stability.

3.2. Sorption of rhodamine B

The adsorption of RB to the MIL-100(Fe) is illustrated in Fig. S7a (SI) as a function of adsorbent dosage. As the adsorbent dosage increased from 0.2 to 2.0 g L^{-1}, the adsorption capacity decreased gradually from 94.4 to 48.4 mg g^{-1}. Meanwhile, as the dosage of the MIL-100(Fe) increased from 0.2 to 2.0 g L^{-1}, the percent removal rate increased from 23.6 to 91.2% due to the increased number of sorption sites available for the RB removal based on an increase of adsorbent dosage.

Fig. S7b (SI) shows the RB adsorption to the MIL-100(Fe) in response to solution pH change. The RB adsorption to the MIL-100(Fe) was unfavorable at pH 2.1 with an adsorption capacity of 19.9 mg g^{-1}. Under a highly acidic condition (pH = 2.1), strong electrostatic repulsion could occur between positively charged MIL-100(Fe) and positively charged RB. It should be noted that RB exists in the protonated form (RBH^+) at pH 1.0–3.0 and in the of zwitterion form ($RB^{\pm}$) at pH > 4.0 (Liu et al., 2016). The RB adsorption to the MIL-100(Fe) became highly favorable at pH 4.1–10.1 with adsorption capacities of 44.0–44.6 mg g^{-1}. At these pH conditions, two mechanisms could play important roles in the adsorption process. One is π-π interaction between the benzene rings of RB and MIL-100(Fe). Another is electrostatic attraction between positively-charged aminoxanthene group of RB and negatively-charged surface of MIL-100(Fe) (Moreno-Villoslada et al., 2006; Liu et al., 2016; Mahmoodi et al., 2019). At pH 11.9, the RB sorption to the MIL-100(Fe) became negligible due to the instability of RB under highly alkaline conditions (Liu et al., 2016).mg (g min)$^{-1}$

The kinetic sorption data are illustrated in Fig. S8a (SI). The kinetic model parameters of pseudo first-order (equation S1), pseudo second-order (equation S2), and Elovich (equation S3) models are listed in Table S3 (SI). The RB adsorption to the MIL-100(Fe) could reach equilibrium at around 6 h. The kinetic data could be well described by the Elovich kinetic model which included two parameters: the initial reaction rate ($\alpha = 13.4$ g mg^{-1}), and the surface coverage ($\beta = 0.196$ mg g^{-1}). The Elovich model indicates that the rate of RB adsorption decreases as the surface coverage on the MIL-100(Fe) increases.

Fig. S8b (SI) presents the equilibrium data, and Table S4 (SI) lists the model parameters of the Langmuir (equation S4), Freundlich (equation S5), and Redlich-Peterson (equation S6) models. The Langmuir model was best fitted to the equilibrium data with the parameter values of $Q_R = 61.9$ L mg^{-1} and $K_L = 0.282$ mg g^{-1}, indicating that the RB adsorption to the MIL-100(Fe) could be described as a monolayer adsorption. The adsorption amount of RB on the surface of the MIL-100(Fe) increased quickly, then gradually slowed down, and approached the saturated adsorption. Similar findings have been reported by Duan et al. (2016) who showed that the RB adsorption to nanoscale NMIL-100(Fe) could be well-described by the Langmuir model. In our study, the maximum RB adsorption capacity of the MIL-100(Fe) was 61.9 mg g^{-1}, which was in the range of values (3.75–918.9 kJ mol^{-1}) reported in the literature (Table S5, SI) (Duan et al., 2016; Liu et al., 2016; Yang et al., 2016; Jin et al., 2018; Zhang et al., 2018; Cui et al., 2019; Li et al., 2019; Jarrah et al., 2020; Navarathna et al., 2020).

The thermodynamic data analyzed with the thermodynamic models (equation S7–S9) are illustrated in Fig. S8c (SI). In Table S6 (SI), the enthalpy (ΔH°) value of 35.4 showed that the RB adsorption to the MIL-100(Fe) was endothermic, increasing with increasing temperature from 20 to 40 °C. The entropy (ΔS°) value of 137 J (K mol)$^{-1}$ demonstrated that the entropy increased at the interface between solid and aqueous phases during the sorption process. The Gibb's free energy (ΔG°) values of (−4.82 to −7.56 kJ mol^{-1}) indicated the spontaneous RB adsorption to the MIL-100(Fe). Our result was in accordance with the studies of other researchers who had observed the endothermic nature of RB adsorption to nanoscale MIL-100(Fe) (Duan et al., 2016) and MIL-53(Fe)-based hybrid adsorbent (MOF-MBC) (Navarathna et al., 2020).

3.3. RSM modeling

The CCD matrix with three input variables is presented in Table 2 along with observed removal rates and predicted values from the RSM modeling. A third order polynomial equation (cubic regression model) was developed from the experimental data in terms of coded factors using RSM follows:

$$y = 16.07A + 15.51B - 5.91C + 0.9722AB + 2.7AC + 7.52BC - 10.14A^2 - 6.61B^2 + 1.8C^2 - 4.12ABC - 10.43A^2B - 31.49A^2C - 32.67AB^2 + 84.32 \quad (7)$$

The ANOVA results of full cubic regression model from RSM are presented in Table S7 (SI). The model *F*-value of 57.9 indicated that the developed cubic model was significant. The lack of fit *p*-value 0.220 implied that the lack of fit was not significant relative to the pure error. The coefficient of determination value (R^2) of 0.98 indicated good correlation between the observed and predicted removal rates. The adequate precision value of 22.5 also demonstrated that the developed RSM model was adequate to describe the RB adsorption to the MIL-100(Fe) (Gadekar and Ahammed, 2019; Mahmoodi et al., 2019). The analysis showed that linear terms (*A*, *B*, *C*), quadratic cross-product term (*BC*), quadratic terms (A^2, B^2), and cubic cross-product terms (A^2B, A^2C, AB^2) had a large effect on the RB adsg L^{-1}orption to the MIL-100(Fe) due to their high *F*-values as well as low *P*-values (<0.05). Fig. S9 (SI) illustrates the diagnostic plot between observed and predicted removal rates, indicating that the prediction of the developed cubic model was acceptable.

Fig. S10 (SI) presents the interaction effect of input variables (pH, adsorbent dose, and initial RB concentration) on the RB removal rate. The removal rate was influenced by the initial RB concentration and adsorbent dose. At pH 4 and adsorbent dose of 2.0, the removal rate decreased from 94.7 to 79.7% with an increase of initial RB concentration from 50 to 150 mg L^{-1}. As the adsorbent dose increased from 1.0 to 3.0 g L^{-1}at pH 4 and initial RB concentration of 100 mg L^{-1}, the removal rate increased from 54.1 to 93.4% (Fig. S10a, S10b). The removal rate was also affected by solution pH. At the adsorbent dose of 2.0 g L^{-1} and initial RB concentration of

100 mg L^{-1}, the removal rate increased from 44.8 to 88.4% with a change of pH from 2 to 6 (Fig. S10c, S10d). As presented in Table 3, the regressor variable of 'A' (pH) had a larger coefficient value (16.1) than 'B' (adsorbent dose) and 'C' (initial RB concentration), indicating that pH had a higher impact on the RB removal rate than other input variables.

3.4. ANN modeling

The structure of the ANN model (Fig. 3) is determined by the number of hidden layers and the number of neurons, which directly affect the performance of ANN (Sharafi et al., 2019). The optimal ANN structure, which provides flexibility to the ANN model and simultaneously avoids over fitting of the model, can be selected by comparing the performances of the various topologies (Gadekar and Ahammed, 2019). In this study, the number of hidden layers and the number of neurons were varied to find the optimal structure for the ANN model (Table S8, SI). Initially, one hidden layer and six hidden layer neurons (3:6:1) were chosen for the network topology. Then, the number of the hidden layer neurons increased to ten (3:10:1) to find the optimal ANN structure. Furthermore, the number of the hidden layers increased to two with the hidden layer neurons varying from six to ten (3:6:6:1–3:10:10:1). Based on the lowest MSE value (0.611) from the validating set, the topology 3:8:1 was selected for the optimal ANN structure, which had three input variables (pH, adsorbent dose, and initial RB concentration), eight neurons in one hidden layer, and one output variable (removal rate).

Based on equation (5), which used the Tansig and Purelin transfer functions, the ANN model with the topology 3:8:1 was developed as follows:

$$y=\sum_{n=1}^{8}\left\{\left(\frac{2}{1+\exp\left[-2\left(\left(A\times w_{A,n}+B\times w_{B,n}+C\times w_{C,n}\right)+b_n\right)\right]}-1\right)\times w_{O,n}\right\}+b_O \tag{8}$$

The values of weights and biases for each layer and neuron in equation (8) are tabulated in Table 3, demonstrating that each neuron has a different combination of the weights and biases. The diagnostic plots between observed and predicted removal rates in all phases of the ANN modeling are presented in Fig. 4. The correlation coefficient (R) value indicated that the developed ANN model [equation (8)] could adequately predict the RB adsorption. The CCD matrix with observed removal rates and predicted values from the developed ANN model is presented in Table 2.

In order to determine the most influential input variable on the RB removal rate, the impacts of three input variables were compared based on the following formulas (Samui et al., 2018; Takdastan et al., 2019):

$$I_i(\%)=\frac{\sum_{n=1}^{8}\left[\frac{|w_{i,n}|\times|w_{O,n}|}{\sum_{i=1}^{3}\left(|w_{i,n}|\times|w_{O,n}|\right)}\right]}{\max\left\{\sum_{n=1}^{8}\left[\frac{|w_{A,n}|\times|w_{O,n}|}{\sum_{i=1}^{3}\left(|w_{i,n}|\times|w_{O,n}|\right)}\right],\ \sum_{n=1}^{8}\left[\frac{|w_{B,n}|\times|w_{O,n}|}{\sum_{i=1}^{3}\left(|w_{i,n}|\times|w_{O,n}|\right)}\right],\ \sum_{n=1}^{8}\left[\frac{|w_{C,n}|\times|w_{O,n}|}{\sum_{i=1}^{3}\left(|w_{i,n}|\times|w_{O,n}|\right)}\right]\right\}}\times 100 \tag{9}$$

$$E_i(\%)=\frac{\sum_{n=1}^{8}\left(|w_{i,n}|\times|w_{O,n}|\right)}{\sum_{i=1}^{3}\left[\sum_{n=1}^{8}\left(|w_{i,n}|\times|w_{O,n}|\right)\right]}\times 100 \tag{10}$$

where I_i is the relative importance of each input variable (%), i is the input variable (A: $i=1$, B: $i=2$, C: $i=3$), n is the number of neuron, $w_{i,n}$ is the weight of input variable 'i' in hidden layer nth neuron, E_i is the effect of each input variables on the output value (%), and $w_{O,n}$ is the weight in output layer nth neuron. The relative importance of the input variable was in the decreasing order of pH (100%) > initial RB concentration (60.7%) > adsorbent dosage (46.7%) (Fig. S11a, SI). The effect of each input variable on the output value was in the order of pH (48.2%) > initial RB concentration (29.3%) > adsorbent dosage (22.5%) (Fig. S11b, SI). The analyses showed that pH was the most important input variable affecting the RB removal rate.

Table 3
Values of weights and biases for each layer and neuron in equation (8).

Neuron (n =)	Hidden layer				Output layer	
	$w_{A,n}$	$w_{B,n}$	$w_{C,n}$	b_n	$w_{O,n}$	b_O
1	2.34	0.875	−1.78	−2.55	0.740	−0.159
2	−1.14	−2.14	−0.969	2.38	0.476	
3	2.78	0.703	−0.689	−1.34	0.619	
4	−3.02	0.897	0.751	0.591	1.04	
5	−1.60	−1.81	−1.80	−1.52	−0.295	
6	−2.52	−0.636	0.780	−1.38	−0.998	
7	0.883	1.45	2.26	2.01	−0.774	
8	1.62	0.447	−2.15	2.95	0.579	

$w_{A,n}$ = weight of input A in hidden layer nth neuron, $w_{B,n}$ = weight of input B in hidden layer nth neuron, $w_{C,n}$ = weight of input C in hidden layer nth neuron, b_n = bias in hidden layer nth neuron, $w_{O,n}$ = weight in output layer nth neuron, b_O = bias in output layer.

3.5. ANN model versus RSM model

In order to further examine the predictability of the developed RSM [equation (7)] and ANN [equation (8)] models for the RB removal rate, additional adsorption experiments were conducted in the experimental conditions, which were not previously included in Table 2. Four experimental conditions were randomly selected in the range of the input variables applied for the CCD matrix using the random number generator (RAND function) in Excel software (version 2016, Microsoft, Redmond, WA, USA). Additional experiments (n = 8) were performed in duplicate for each experimental condition. The observed removal rates from the additional

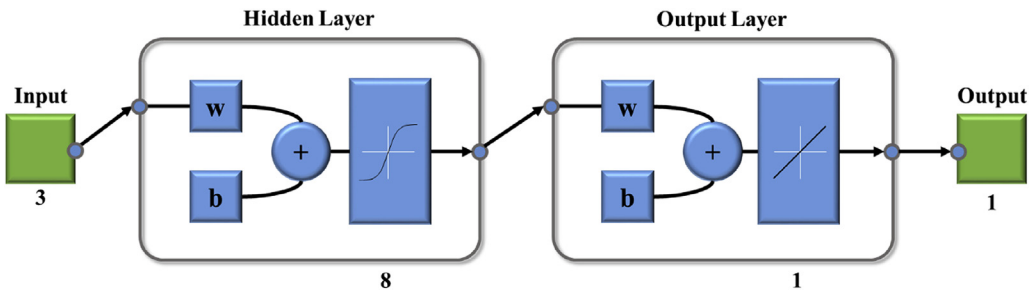

Fig. 3. Artificial neuron network (ANN) structure with a topology of 3:8:1.

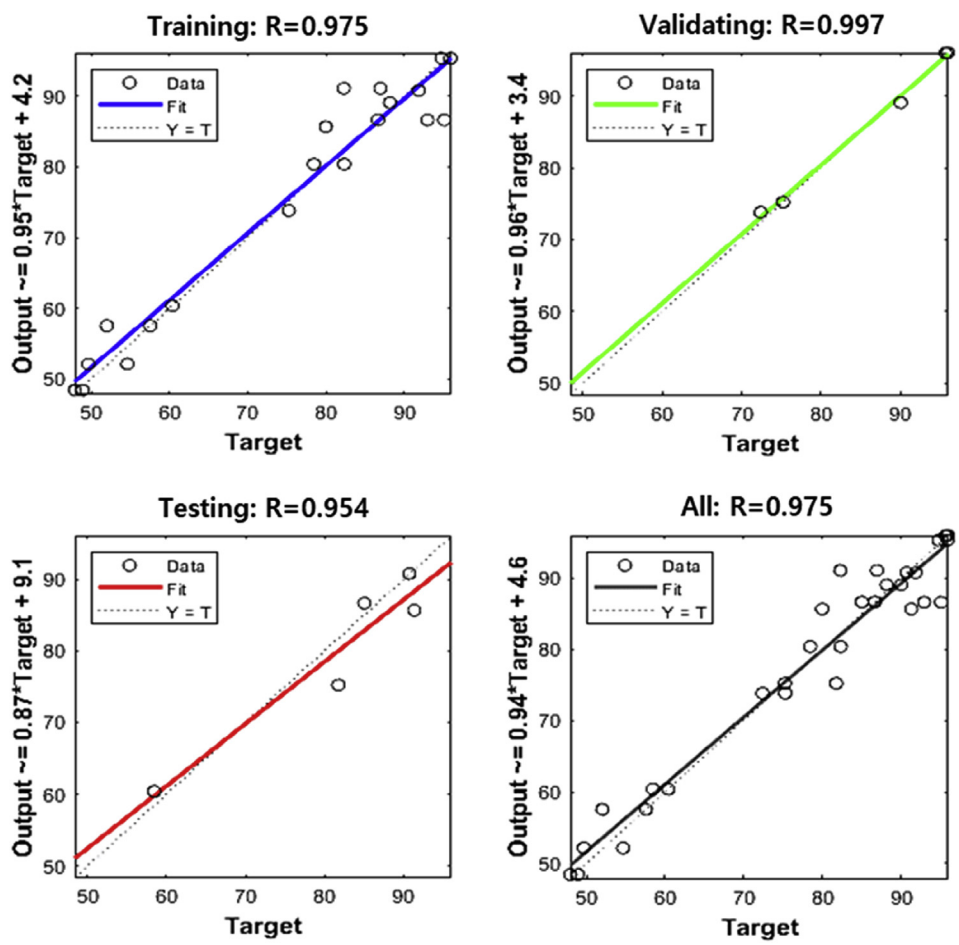

Fig. 4. Diagnostic plots between observed and predicted removal rates in the ANN modeling.

Table 4
Observed removal rates from additional adsorption experiments along with predicted values from the RSM and ANN models.

Exp.	Input variable			Removal rate (%)		
	A	B	C	Observed	Predicted-RSM	Predicted-ANN
A1	3.1	1.5	107	60.6	64.7	59.1
A2	3.0	2.0	60	87.8	92.9	85.5
A3	4.1	2.7	72	95.0	92.8	95.6
A4	4.2	2.5	121	75.0	91.2	86.8
A5	3.1	1.5	107	68.7	64.7	59.1
A6	3.0	2.0	60	88.6	92.9	85.5
A7	4.1	2.7	72	96.1	92.8	95.6
A8	4.2	2.5	121	82.5	91.2	86.8

experiments along with predicted values from the RSM and ANN models are presented in Table 4. Diagnostic plots between observed and predicted removal rates (Fig. 5) demonstrated that the ANN model (R^2 = 0.821) had better predictability for the additional adsorption experiments than the RSM model ($R^2 = 0.733$). Based on the developed ANN model [equation (8)], the optimal condition for the RB adsorption was determined using the MATLAB nntool. The RB removal rate of >99.9% was obtained at the optimum condition of pH 5.3, adsorbent dose 2.0 g L^{-1}, and initial RB concentration 73 mg L^{-1} (Table S9, SI) in the range of the input variables applied for the CCD matrix.

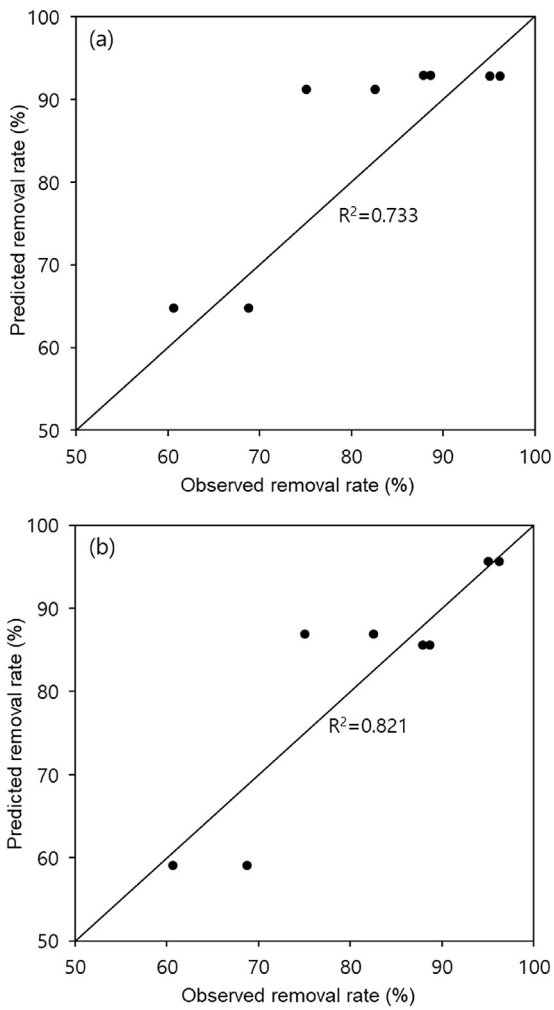

Fig. 5. Diagnostic plots between observed and predicted removal rates from additional experiments: (a) RSM model and (b) ANN model.

4. Conclusions

The synthetic MIL-100(Fe) was applied as an adsorbent for the RB removal from aqueous solutions. RSM and ANN were employed to model and optimize the RB removal by the MIL-100(Fe). The cubic regression model was developed in the RSM modeling. The ANOVA test demonstrated that the developed RSM model was adequate to describe the RB adsorption. The ANN model with the topology of 3:8:1 was developed to adequately model the RB adsorption. The comparative study with additional adsorption data showed that the ANN model had better predictability than the RSM model. The ANN model predicted the optimum RB removal rate (>99.9%) at pH 5.3, adsorbent dose 2.0, and initial RB concentration 73 g L^{-1}. In addition, pH was the most important input variable affecting the RB removal rate. This study demonstrated that the MIL-100(Fe) could be effectively used to remove RB from aqueous solutions. In addition, the developed ANN model could be employed to model and optimize the RB adsorption to the MIL-100(Fe). This study was, however, limited to the application of MIL-100(Fe) to the RB removal in synthetic solutions. Further experiments and modeling studies are required for dye removal by MIL-100(Fe) in industrial wastewater.

Credit author statement

Ho-Young Jang, Investigation, Writing - original draft. Jin-Kyu Kang, Conceptualization. Jeong-Ann Park, Methodology. Seung-Chan Lee, Software. Song-Bae Kim, Supervision, Writing-reviewing and editing

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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References

Abbasi, A.R., et al., 2017. Efficient removal of crystal violet and methylene blue from wastewater by ultrasound nanoparticles Cu-MOF in comparison with mechanosynthesis method. Sonochem 37, 182–191.
Adeyemo, A.A., et al., 2012. Metal organic frameworks as adsorbents for dye adsorption: overview, prospects and future challenges. Toxicol. Environ. Chem. 94, 1846–1863.
Apostolopoulou, M., et al., 2020. Mapping and holistic design of natural hydraulic lime mortars. Cement Concr. Res. 136, 106167.
Askari, H., et al., 2017. Rapid and high-capacity ultrasonic assisted adsorption of ternary toxic anionic dyes onto MOF-5-activated carbon: artificial neural networks, partial least squares, desirability function and isotherm and kinetic study. Sonochem 37, 71–82.
Aslam, S., et al., 2017. In situ one-step synthesis of Fe_3O_4@MIL-100(Fe) core-shells for adsorption of methylene blue from water. J. Colloid Interface Sci. 505, 186–195.
Ayati, A., et al., 2016. Emerging adsorptive removal of azo dye by metal-organic frameworks. Chemosphere 160, 30–44.
Baldev, E., et al., 2013. Degradation of synthetic dye, rhodamine B to environmentally non-toxic products using microalgae. Colloids Surf., B 105, 207–214.
Baziar, M., et al., 2017. MWCNT-Fe_3O_4 as a superior adsorbent for microcystins LR removal: investigation on the magnetic adsorption separation, artificial neural network modeling, and genetic algorithm optimization. J. Mol. Liq. 241, 102–113.
Boudechiche, N., et al., 2017. Valorization of crataegus azarolus stones for the removal of textile anionic dye by central composite rotatable design using cubic model: optimization, isotherm, and kinetic studies. Environ. Sci. Pollut. Res. 24, 19609–19623.
Cai, J., et al., 2016. Selective adsorption of arsenate and the reversible structure transformation of the mesoporous metal-organic framework MIL-100(Fe). Phys. Chem. Chem. Phys. 18, 10864–10867.
Cui, W., et al., 2019. Gel-like ZnO/Zr-MOF(bpy) nanocomposite for highly efficient adsorption of Rhodamine B dye from aqueous solution. J. Phys. Chem. Solid. 134, 165–175.
Duan, S., et al., 2016. HF-free synthesis of nanoscale metal-organic framework NMIL-100(Fe) as an efficient dye adsorbent. ACS Sustain. Chem. Eng. 4, 3368–3378.
Esfandiari, K., et al., 2017. Using artificial neural network and ideal adsorbed solution theory for predicting the CO_2/CH_4 selectivities of metal-organic frameworks: a comparative study. Ind. Eng. Chem. Res. 56, 14610–14622.
Esrafili, L., et al., 2019a. Selective detection and removal of mercury ions by dual-functionalized metal-organic frameworks: design-for-purpose. New J. Chem. 43, 18079–18091.
Esrafili, L., et al., 2019b. The targeted design of dual-functional metal-organic frameworks (DF-MOFs) as highly efficient adsorbents for Hg^{2+} ions: synthesis for purpose. Dalton Trans. 48, 17831–17839.
Fan, J., et al., 2018. Adsorption and biodegradation of dye in wastewater with Fe_3O_4@MIL-100(Fe) core-shell bio-nanocomposites. Chemosphere 191, 315–323.
Fang, Y., et al., 2018. Effect of mineralizing agents on the adsorption performance of metal-organic framework MIL-100(Fe) towards chromium(VI). Chem. Eng. J. 337, 532–540.

Feng, M., et al., 2018. Water-stable metal-organic frameworks for aqueous removal of heavy metals and radionuclides: a review. Chemosphere 209, 783–800.
Furukawa, H., et al., 2013. The chemistry and applications of metal-organic frameworks. Science 341, 1230444.
Gadekar, M.R., et al., 2019. Modelling dye removal by adsorption onto water treatment residuals using combined response surface methodology-artificial neural network approach. J. Environ. Manag. 231, 241–248.
Ghaedi, A.M., et al., 2017. Applications of artificial neural networks for adsorption removal of dyes from aqueous solution: a review. Adv. Colloid Interface Sci. 245, 20–39.
Ghaedi, M., et al., 2015. Artificial neural network-genetic algorithm based optimization for the adsorption of phenol red (PR) onto gold and titanium dioxide nanoparticles loaded on activated carbon. J. Ind. Eng. Chem. 21, 587–598.
Guesh, K., et al., 2017. Sustainable preparation of MIL-100(Fe) and its photocatalytic behavior in the degradation of methyl orange in water. Cryst. Growth Des. 17, 1806–1813.
Haque, E., et al., 2011. Adsorptive removal of methyl orange and methylene blue from aqueous solution with a metal-organic framework material, iron terephthalate (MOF-235). J. Hazard Mater. 185, 507–511.
Huo, S.H., et al., 2012. Metal-organic framework MIL-100(Fe) for the adsorption of malachite green from aqueous solution. J. Mater. Chem. 22, 7449–7455.
Jahed Armaghani, D., et al., 2014. Indirect measure of shale shear strength parameters by means of rock index tests through an optimized artificial neural network. Measurement 55, 487–498.
Jahed Armaghani, D., et al., 2017. Developing a hybrid PSO-ANN model for estimating the ultimate bearing capacity of rock-socketed piles. Neural Comput. Appl. 28, 391–405.
Jarrah, A., et al., 2020. Encapsulation of $K_6P_2W_{18}O_{62}$ into magnetic nanoporous Fe_3O_4/MIL-101(Fe) for highly enhanced removal of organic dyes. J. Solid State Chem. 285, 121264.
Jin, L., et al., 2018. Nickel nanoparticles encapsulated in porous carbon and carbon nanotube hybrids from bimetallic metal-organic-frameworks for highly efficient adsorption of dyes. J. Colloid Interface Sci. 509, 245–253.
Kökçam-Demir, U., et al., 2020. Coordinatively unsaturated metal sites (open metal sites) in metal-organic frameworks: design and applications. Chem. Soc. Rev. 49, 2751–2798.
Lee, J.G., et al., 2017. Supersonically sprayed gas- and water-sensing MIL-100(Fe) films. J. Alloys Compd. 722, 996–1001.
Li, Y., et al., 2019. Trace pyrolyzed ZIF-67 loaded activated carbon pellets for enhanced adsorption and catalytic degradation of rhodamine B in water. Chem. Eng. J. 375, 122003.
Liu, H., et al., 2016. Synthesis and characterization of magnetic metal-organic framework for the adsorptive removal of rhodamine B from aqueous solution. J. Ind. Eng. Chem. 34, 278–285.
Liu, Z., et al., 2018. Synthesis of magnetic orderly mesoporous α–Fe_2O_3 nanocluster derived from MIL-100(Fe) for rapid and efficient arsenic(III,V) removal. J. Hazard Mater. 343, 304–314.
Ma, S., et al., 2019. Hydrogen purification layered bed optimization based on artificial neural network prediction of breakthrough curves. Int. J. Hydrogen Energy 44, 5324–5333.
Mahmoodi, N.M., et al., 2019. Novel magnetic amine functionalized carbon nanotube/metal-organic framework nanocomposites: from green ultrasound-assisted synthesis to detailed selective pollutant removal modelling from binary systems. J. Hazard Mater. 368, 746–759.
Mohammadifard, Z., et al., 2019. Heterogeneous photo-fenton degradation of formaldehyde using MIL-100(Fe) under visible light irradiation. Environ. Pollut. 251, 783–791.
Moreno-Villoslada, I., et al., 2006. pH dependence of the interaction between Rhodamine B and the water-soluble poly(sodium 4-styrenesulfonate). J. Phys. Chem. B 110, 11809–11812.
Nair, A.T., et al., 2014. The use of response surface methodology for modelling and analysis of water and wastewater treatment processes: a review. Water Sci. Technol. 69, 464–478.
Navarathna, C.M., et al., 2020. Rhodamine B adsorptive removal and photocatalytic degradation on MIL-53-Fe-MOF/magnetic magnetite/biochar composites. J. Inorg. Organomet. Polym. 30, 214–229.
Nehra, M., et al., 2019. Metal organic frameworks MIL-100(Fe) as an efficient adsorptive material for phosphate management. Environ. Res. 169, 229–236.
Parsazadeh, N., et al., 2018. Preparation and characterization of monoliths HKUST-1 MOF via straightforward conversion of $Cu(OH)_2$-based monoliths and its application for wastewater treatment: artificial neural network and central composite design modeling. New J. Chem. 42, 10327.
Saldana-Robles, A., et al., 2014. Optimization of the operating parameters using RSM for the fenton oxidation process and adsorption on vegetal carbon of MO solutions. J. Ind. Eng. Chem. 20, 848–857.
Samadi-Maybodi, A., et al., 2020. Removal of sarafloxacin from aqueous solution by a magnetized metal-organic framework; artificial neural network modeling. Polyhedron 179, 114342.
Samui, P., et al., 2018. Integrating Disaster Science and Management: Global Case Studies in Mitigation and Recover. Elsevier Science Publishing Co Inc., Cambridge, MA, USA, pp. 47–49.
Seo, P.W., et al., 2016. Adsorptive removal of pharmaceuticals and personal care products from water with functionalized metal-organic frameworks: remarkable adsorbents with hydrogen-bonding abilities. Sci. Rep. 6, 34462.
Sharafi, K., et al., 2019. Phenol adsorption on scoria stone as adsorbent – application of response surface method and artificial neural networks. J. Mol. Liq. 274, 699–714.
Sharma, N., et al., 2015. WEDM process variables investigation for HSLA by response surface methodology and genetic algorithm. Eng. Sci. Technol. Int. J. 18, 171–177.
Takdastan, A., et al., 2019. Alkali modified oak waste residues as a cost-effective adsorbent for enhanced removal of cadmium from water: isotherm, kinetic, thermodynamic and artificial neural network modeling. J. Ind. Eng. Chem. 78, 352–363.
Tan, F., et al., 2015. Facile synthesis of size-controlled MIL-100(Fe) with excellent adsorption capacity for methylene blue. Chem. Eng. J. 281, 360–367.
Tehrani, A.A., et al., 2017. Urea metal-organic frameworks for nitro-substituted compounds sensing. Inorg. Chem. 56, 1446–1454.
Torad, N.L., et al., 2014. Direct synthesis of MOF-derived nanoporous carbon with magnetic Co nanoparticles toward efficient water treatment. Small 10, 2096–2107.
Wang, R., et al., 2017. Highly efficient MOF-based self-propelled micromotors for water purification. RSC Adv. 7, 42462–42467.
Wei, F., et al., 2018. Comparison study on the adsorption capacity of rhodamine B, Congo red, and orange II on Fe-MOFs. Nanomaterials 8, 248.
Yang, C., et al., 2016. Indium-based metal-organic framework/graphite oxide composite as an efficient adsorbent in the adsorption of rhodamine B from aqueous solution. J. Alloys Compd. 687, 804–812.
Yilmaz, E., et al., 2016. Synthesis, characterization of a metal organic framework: MIL-53(Fe) and adsorption mechanisms of methyl red onto MIL-53(Fe). J. Taiwan Inst. Chem. Eng. 65, 323–330.
Zhan, X.Q., et al., 2018. Ligands-coordinated Zr-based MOF for wastewater treatment. Nanomaterials 8, 655.
Zhang, J., et al., 2018. Rapid and selective adsorption of cationic dyes by a unique metal-organic framework with decorated pore surface. Appl. Surf. Sci. 440, 1219–1226.
Zhang, W., et al., 2019. Removal of antimonite and antimonite from water using Fe-based metal-organic frameworks: the relationship between framework structure and adsorption performance. J. Environ. Sci. 86, 213–224.
Zhao, X., et al., 2015. Synthesis of magnetic metal-organic framework (MOF) for efficient removal of organic dyes from water. Sci. Rep. 5, 11849.
Zhou, H.C., et al., 2014. Metal-organic frameworks (MOFs). Chem. Soc. Rev. 43, 5415–5418.
Zhu, T.T., et al., 2016. Encapsulation of tungstophosphoric acid into harmless MIL-101(Fe) for effectively removing cationic dye from aqueous solution. RSC Adv. 6, 81622–81630.